

SAM ALBO

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Education

McMaster University

2027

Double Major in Computer Science and Economics (B.A.)

Hamilton, Ontario

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Software Development, Computer Architecture, Software Design, Databases, Linear Algebra, Intermediate Economics, Statistics

Technical Skills

Languages: Python, C, Java, SQL, R, HTML, CSS

Developer Tools: Visual Studio Code, PyCharm, GitHub

Technologies/Frameworks: Linux, Git, Docker, Power BI, Excel

Work Experience

Data Analyst Intern

Sept 2025 – Dec 2025

Support Ontario Youth

Toronto, Ontario

- **Extracted, transformed, and manipulated** data from multiple disparate sources to facilitate ****diagnostic analytics**** and support high-level organizational reporting.
- **Cleaned, validated, and standardized** large-scale datasets, identifying and resolving inconsistencies to improve data accuracy and strengthen ****data-driven decision-making****.
- Performed **complex data analysis** of internal accounting workflows to develop **23 standardized finance policies**, ensuring a uniform framework for managerial review and approval.
- **Categorized and processed** high-volume financial records using **Dext**, verifying accuracy and assigning expense codes to maintain reliable information within **relational databases**.

Technical Projects

Vanguard-AI: Clinical Predictive Risk Engine | [Repo](#) | *Python, Scikit-Learn, NumPy, Matplotlib, Pandas* Jan 2026

- **Developed** a high-performance clinical decision support system utilizing a **First-Principles Numerical ODE Solver** to model non-linear drug accumulation (PK/PD) over multi-dose cycles.
- **Implemented** a time-stepped **Euler-Method Numerical Integrator** to solve complex differential equations, simulating physiological drug "stacking" effects with high precision.
- **Trained and optimized** a **Random Forest Classifier** on 2,000+ synthetic patient records to predict Acute Kidney Injury (AKI) risk, achieving high accuracy in identifying high-risk physiological patterns.
- **Architected** an automated risk-stratification pipeline that visualizes therapeutic window violations using **Matplotlib**, enabling real-time clinical safety flagging.

McMaster Enrolment Automation Engine | [Repo](#) | *Python, Selenium, Automation, Chrome WebDriver* July 2025

- **Engineered** a robust web-automation engine using **Python and Selenium** to streamline course registration for 50+ concurrent users, maintaining a **100% success rate** during high-traffic periods.
- **Implemented** dynamic wait-states and **asynchronous element selection** to handle non-deterministic server response times, reducing registration latency by **90%**.
- **Designed** an automated scheduling module for timed execution, bypassing manual bottlenecks and ensuring precise, sub-second interaction with the university's relational database interface.
- **Integrated** comprehensive **exception handling and logging** to ensure system resilience against network timeouts and UI changes, ensuring zero-fault execution.

Leadership Experience

Lead Counselor — Program Coordinator

Sept 2023 – Dec 2023

Breaking Barriers – Oakville Soccer Club

Oakville, Ontario

- **Directed and facilitated** inclusive programs for 30+ participants, utilizing ****adaptive leadership**** to customize workflows for diverse cognitive and physical requirements.
- **Coordinated cross-functional communication** between staff and volunteers to iterate on session designs, ensuring 100% engagement through ****accessibility-focused**** problem solving.